

Lecture 15

ChE 467/596-012 Polymer Rheology

Saad A Khan

Gels and Networks

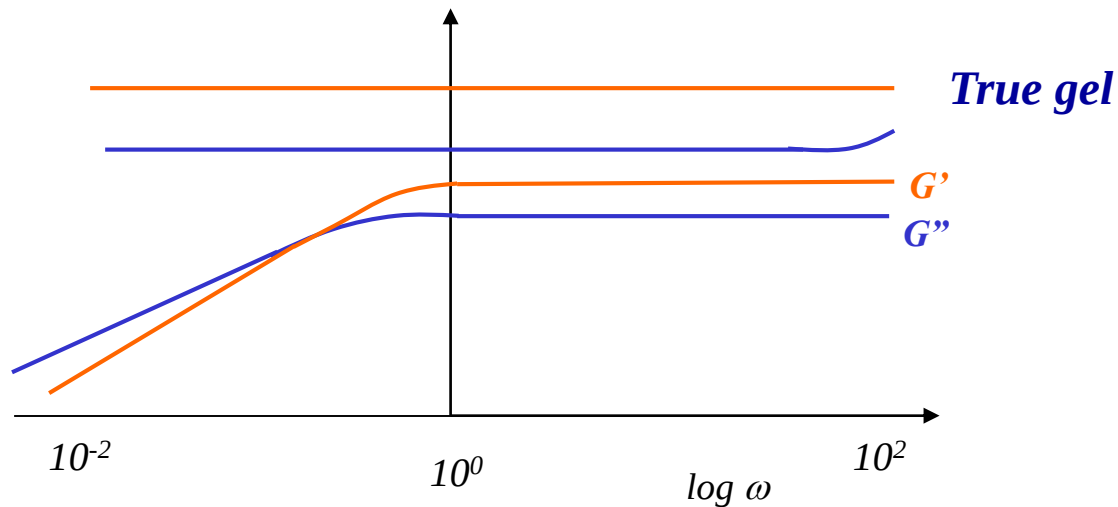
**Probing microstructure, state and kinetics
through rheology**

Gels?

What is a Gel?

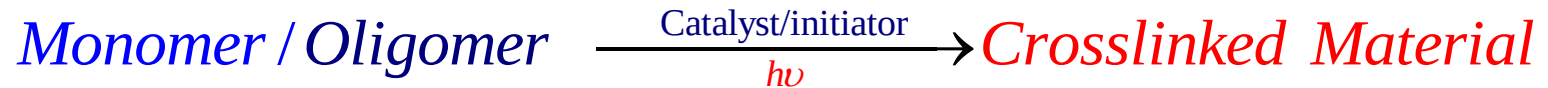
How can you tell if a sample is a gel?

What is the relaxation time of a gel?



Gels and networks

Chemically x-linked systems



Physical Gels (reversible)

Temperature, light, hydrogen bonds

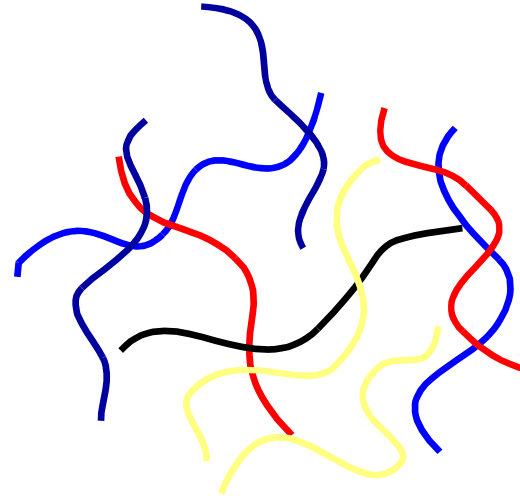
What is a Gel?

3-Dimensional network spanning the sample

Gels and networks, cont'd

Polymer chains:

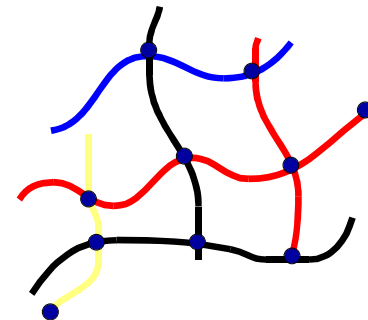
Chains can move/wiggle around through “reptation”



Gels:

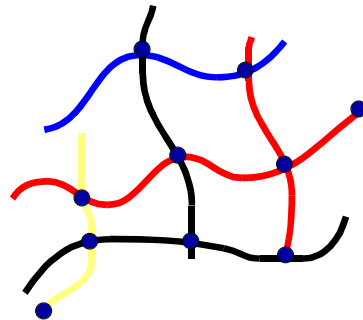
The polymer chains are constrained

Junctions can be fixed chemically or physically



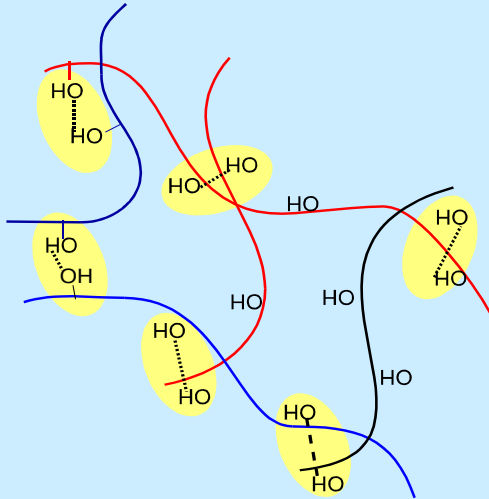
Chemically x-linked gels

Fix Junctions chemically $\Rightarrow$ Covalent bonds
 $\Rightarrow$ Chemically x-linked gels/system
(Chemically cured system/thermosets)

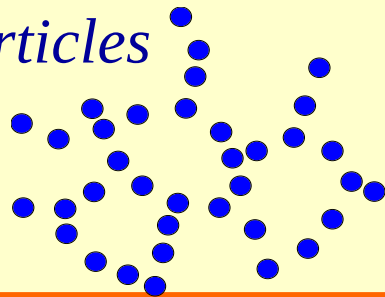


Physical Gel-Examples

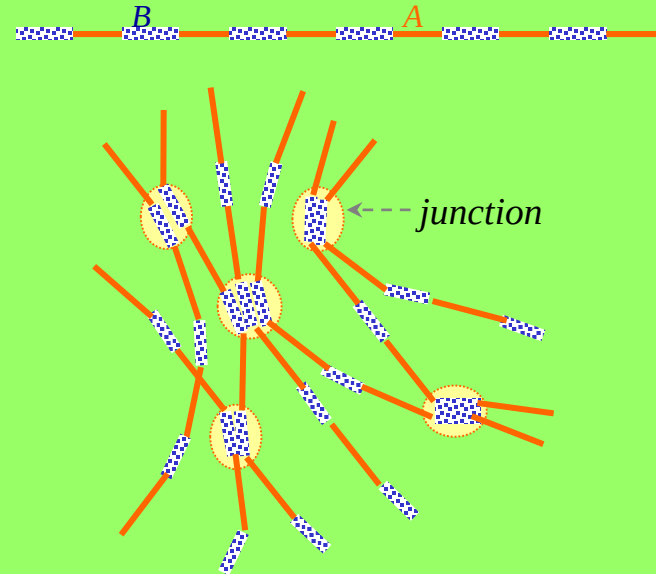
Systems with Hydrogen bonds



Colloidal Particles



Block Copolymers

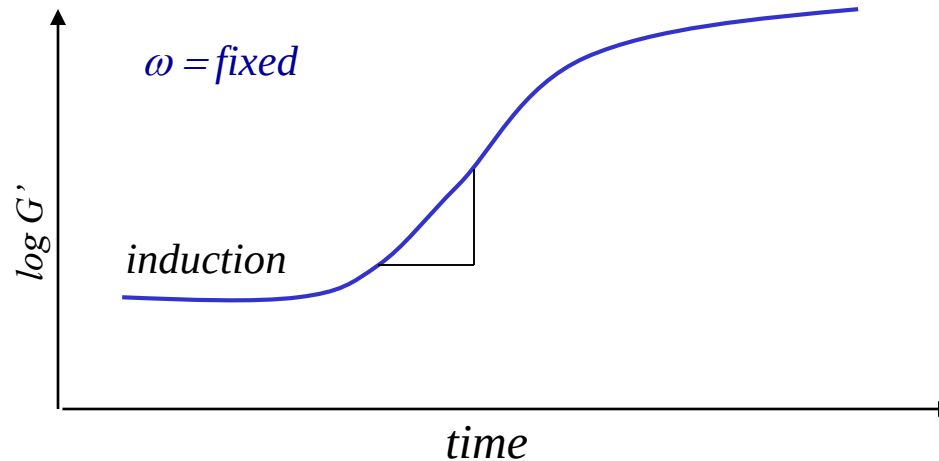


- *Ionic interactions*
- *Hydrophobic interactions*

Critical Issues

1. *What is a good way to monitor gelation/x-linking rheologically?*
2. *How can one differentiate between a gel and a sol/solution?*
3. *How can one get the gel point?*

Monitoring Gelation Kinetics

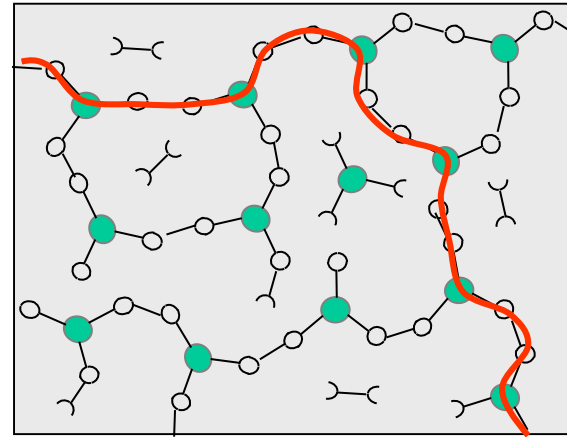
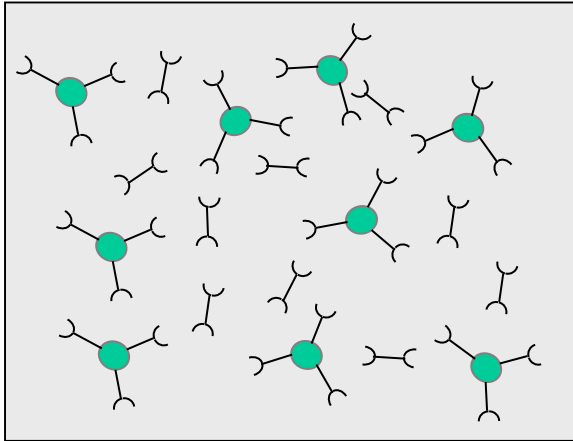


$$G'(t) = n(t) kT + G_{en}$$

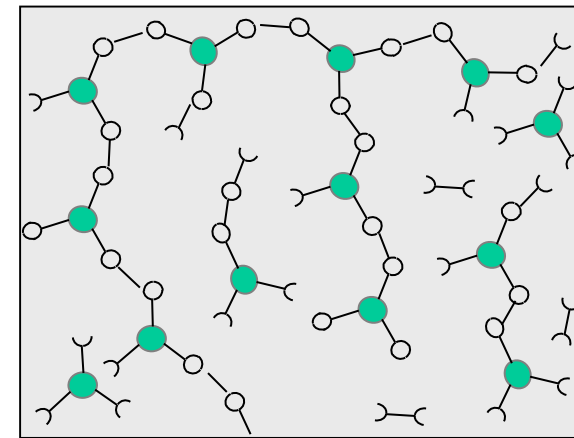
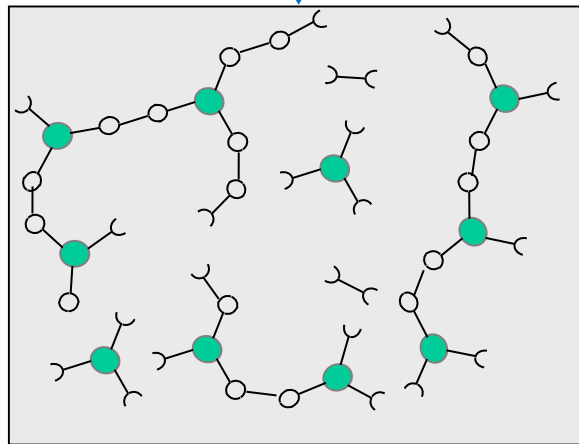
Equivalent to # x-links

$\frac{dG'}{dt}$ is used to relate to kinetics

Microstructure Evolution During X-linking

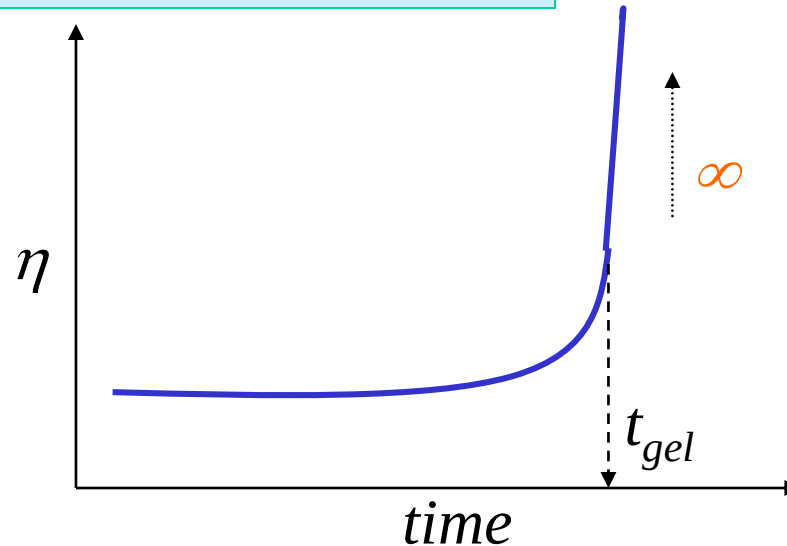


Gel point



Gel point detection – Approach 1

1. Viscosity versus time behavior

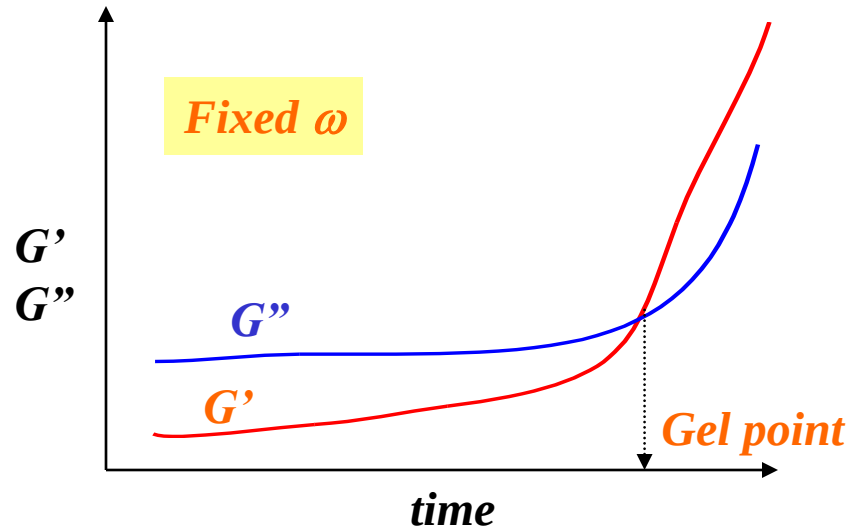


Time at which $\eta \rightarrow \infty \Rightarrow$ gelation time $\rightarrow$ Gelation process

Any Problem? shear can interfere with crosslinking/gelation

Gel point detection-Approach 2

2. G' , G'' versus time behavior



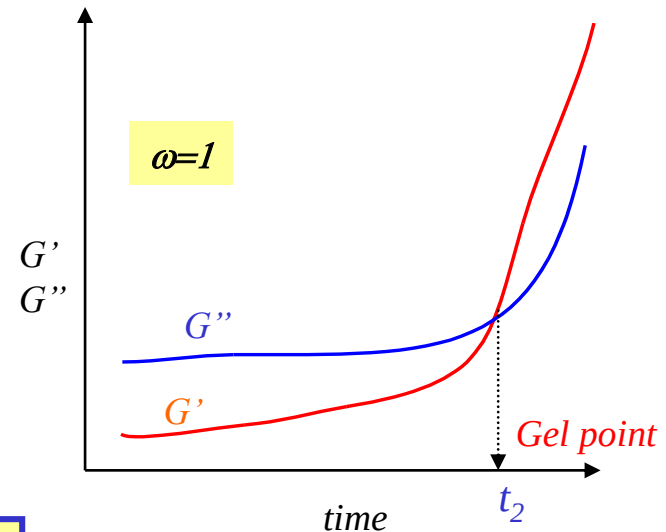
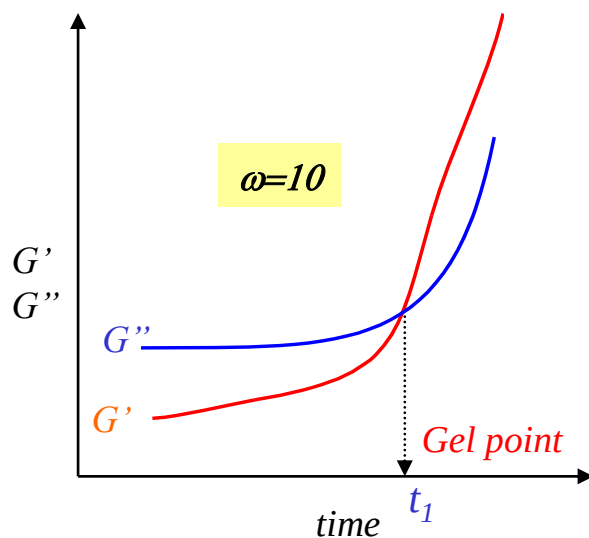
Intersection where $G'=G'' \Rightarrow$ Gel point

Problem: *Do we get same point for different frequencies?*

Gel point detection – Approach 2, cont'd

2. G' , G'' versus time behavior

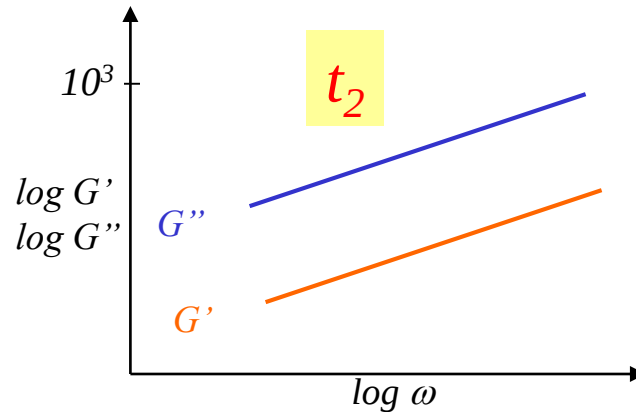
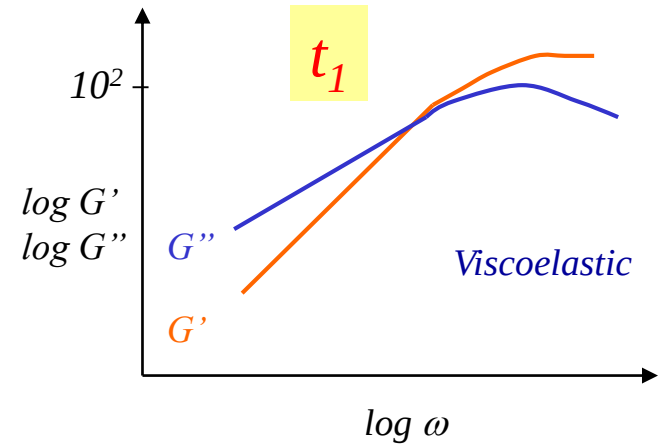
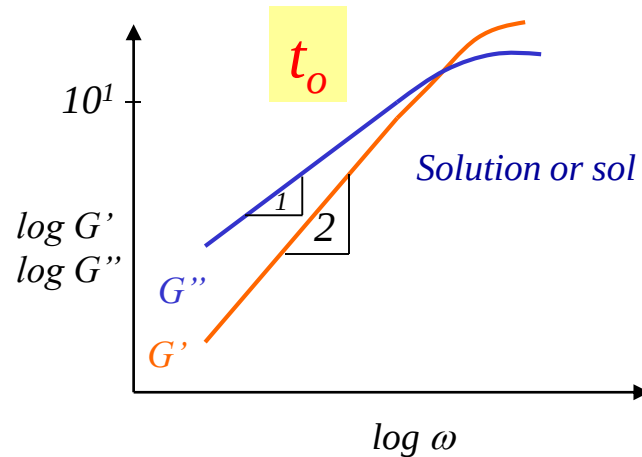
- The technique is ok if gel point is frequency independent. But often this may not be the case.



$$t_1 \neq t_2$$

Gel point detection – Approach 3

3. G' , G'' versus frequency at different time intervals



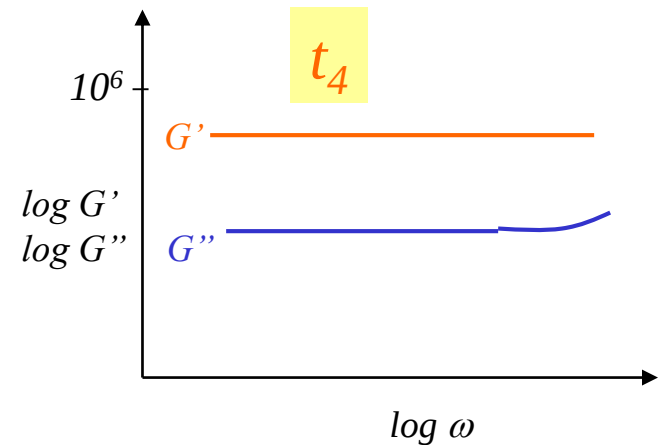
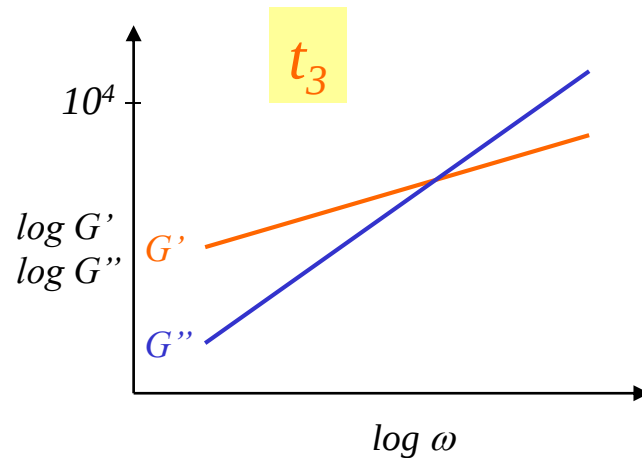
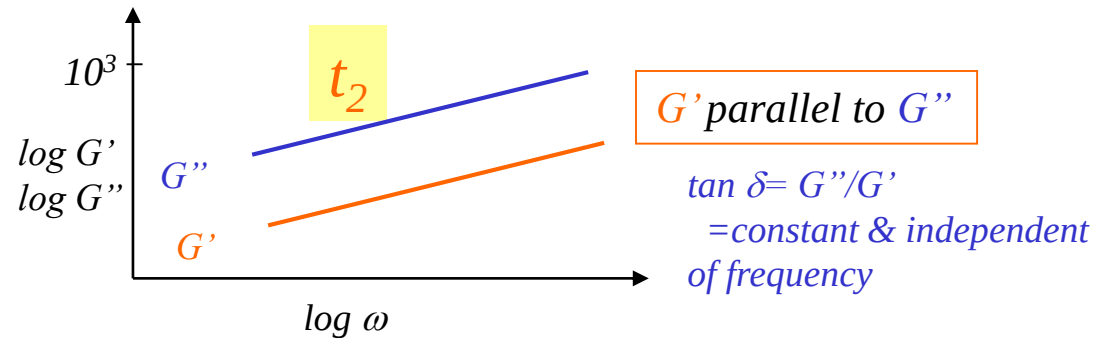
G' parallel to G''

Gel Point

Winter-Chambon
criteria

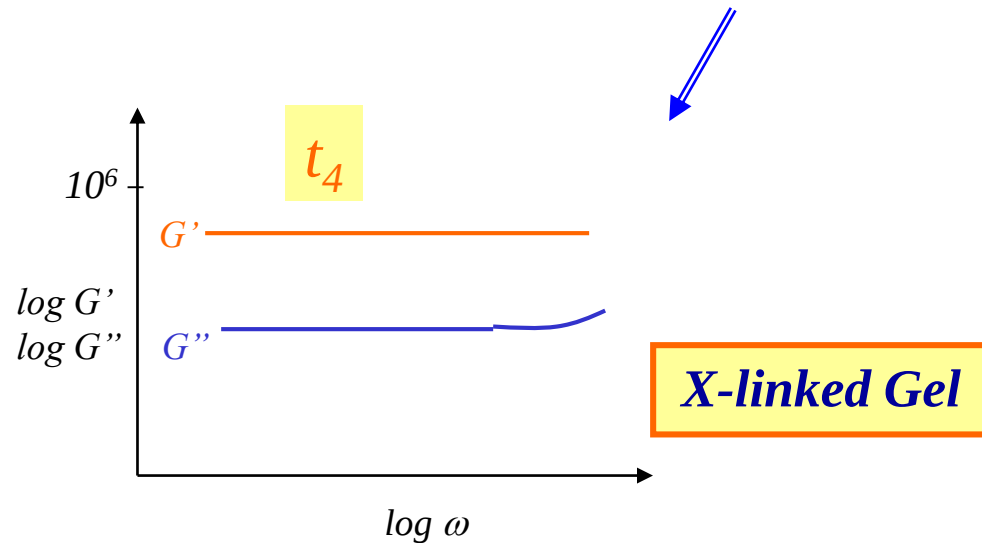
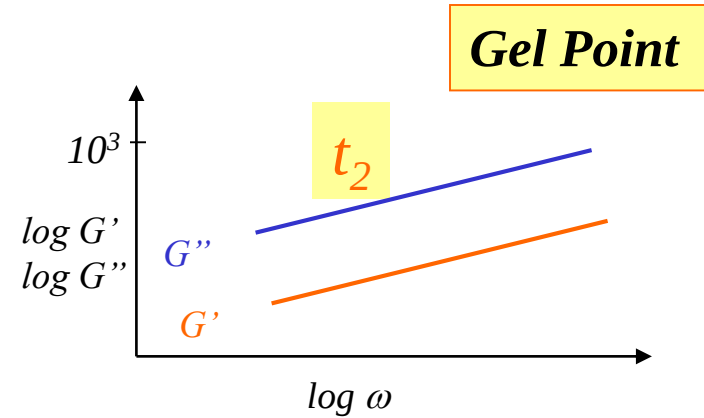
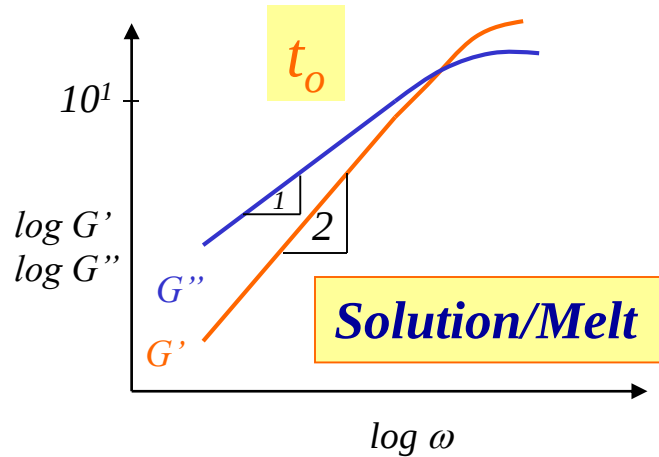
Gelation & Gel point detection

Gel Point



$G' \gg G''$
 G', G'' frequency independent
 = **Fully x-linked gel**

Recap- Gelation & Gel point detection

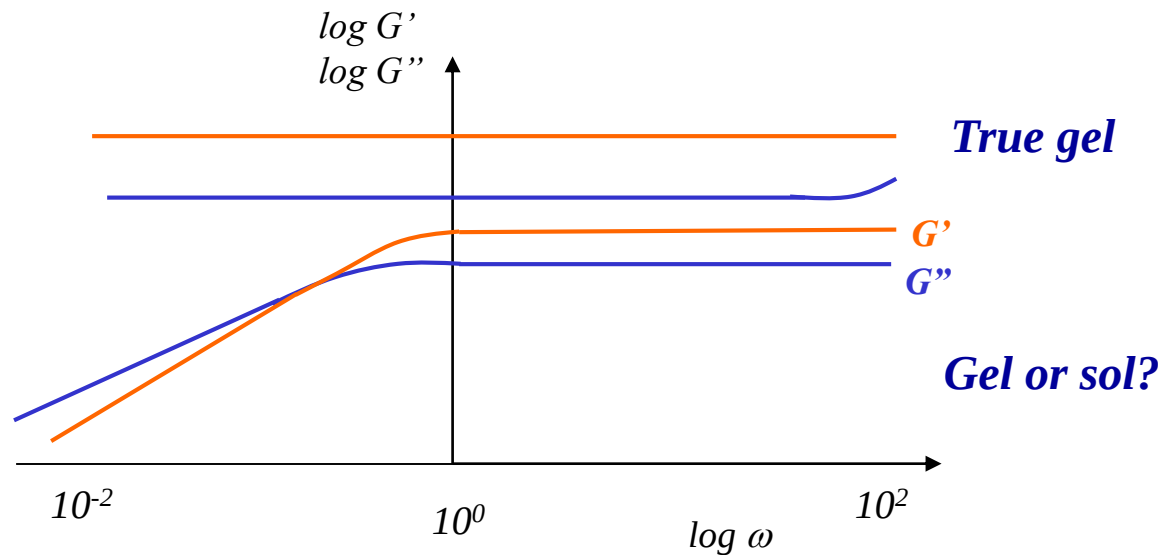


At the Gel Point

- *Molecular weight M_w goes to infinite*
- *System loses solubility*
- *Zero shear viscosity goes to infinite*
- *Equilibrium Modulus is zero and starts to rise to a finite number beyond the gel point*

Exercise – Is this a gel or solution?

Go to lower frequencies to insure G' and G'' are frequency independent



Back to Gel point analysis

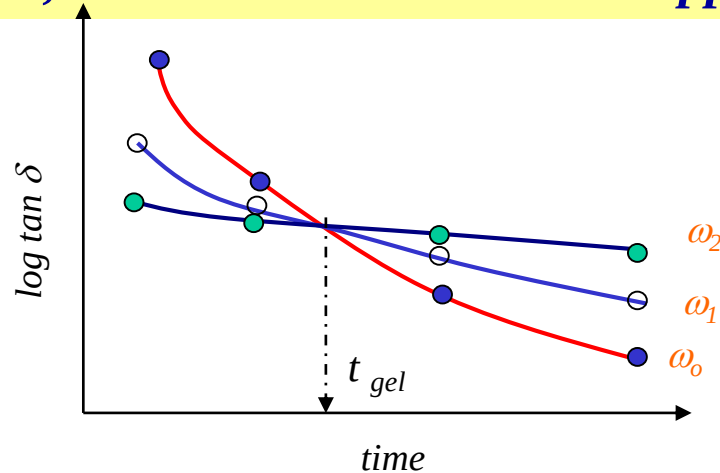
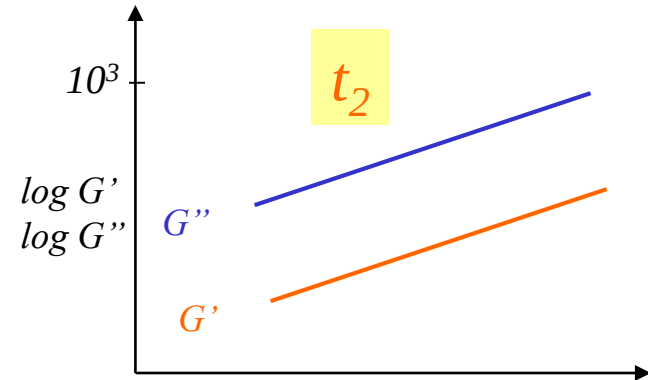
Concern:

*What if we have large enough time interval
so we miss gelation point (t_2 behavior)*

Solution:

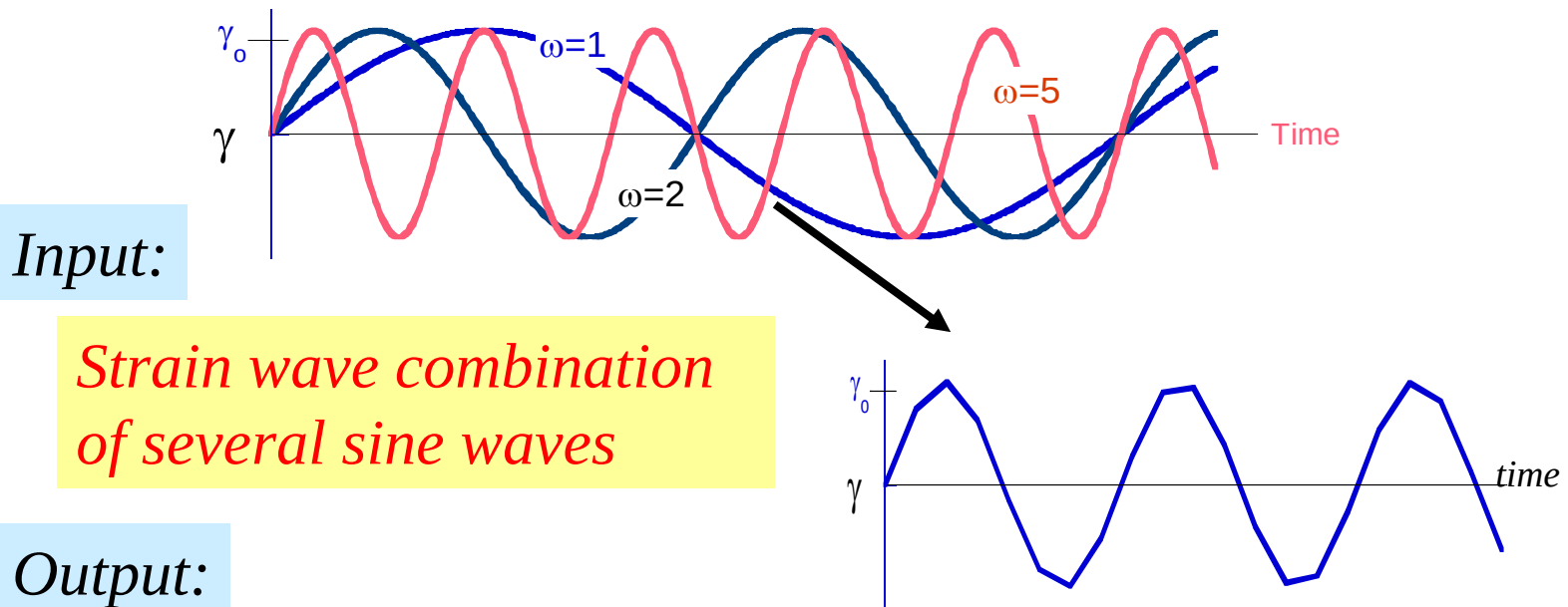
Plot $\tan \delta$ versus time

For fast reactions, in addition use Multiwave approach



Multiwave Approach

Combine several frequencies/oscillation into one.
Fourier Transform Mechanical Spectroscopy



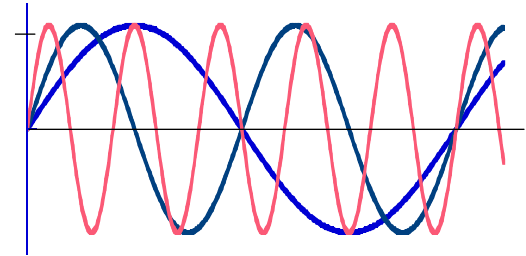
Complex torque signal $\xrightarrow{\text{Fourier transform}}$

G', G'' for several frequencies from one data point

Multiwave Approach, cont'd

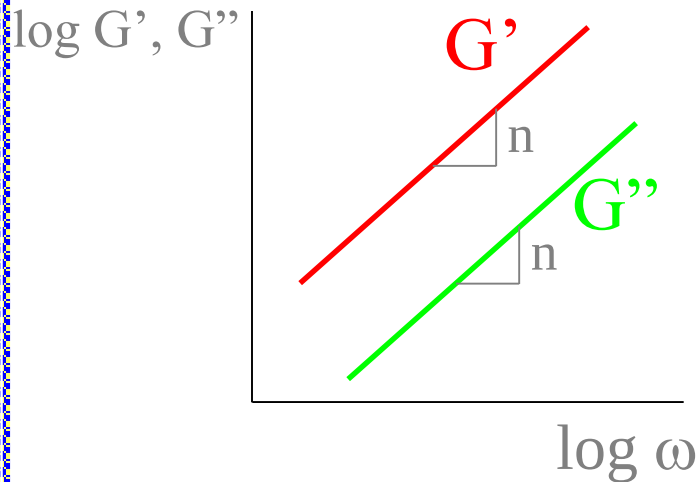
Advantages:

- *Effective in fast reacting system*
- *Only one data point gives several frequencies*
 - Can get G' , G'' versus ω from 1 data point*
- *Can monitor x-linking quickly*



Criterion for Gel Point

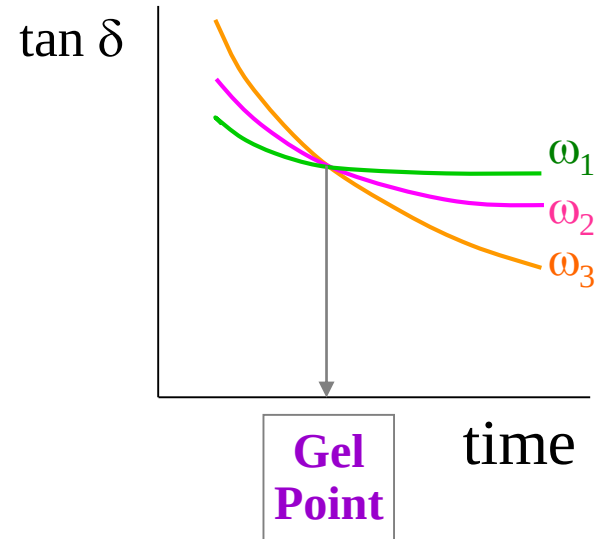
Frequency sweeps at gel point



Elastic Modulus $G' \sim \omega^n$

Viscous Modulus $G'' \sim \omega^n$

Loss tangents at different frequencies



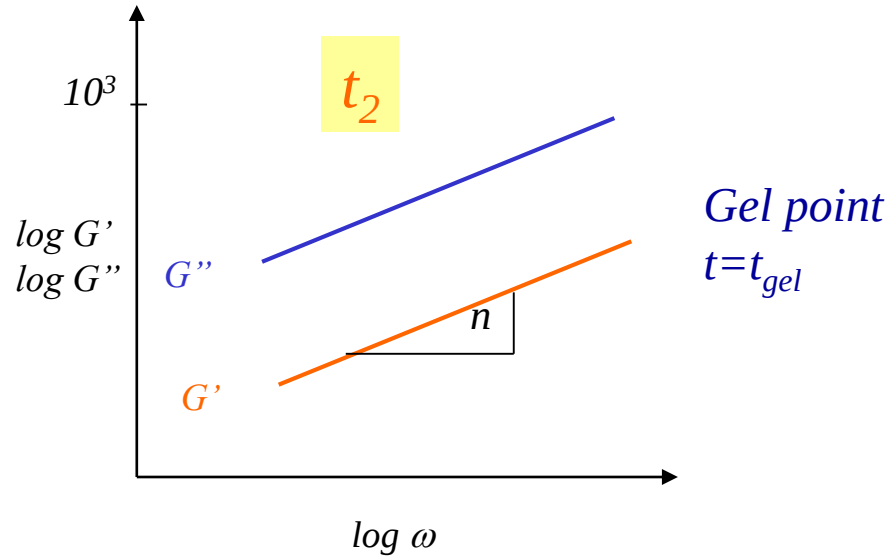
$$\tan \delta = \frac{G''}{G'} = \tan \left(\frac{n\pi}{2} \right)$$

(**independent** of frequency)

Gel & Network-Fractals

Concept of fractals:

Gel point data $\Rightarrow$ *Fractal Dimension* $\Rightarrow$ *Mesh size*



n is related to fractal dimension

Gel&Network-Fractals, cont'd

n is related to fractal dimension:

$$n = \frac{d(d + 2 - 2d_f)}{2(d + 2 - d_f)}$$

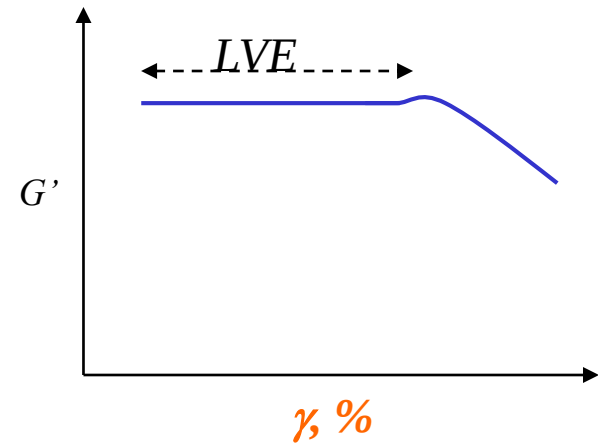
- *d_f is the fractal dimension (can be fractional)*
- *d is the dimension of the sample (=3)*

Multiwave Approach, cont'd

Pick frequency ω_1 : other frequencies have to be multiples of ω_1

Input is superposition of multiple oscillatory strains: $\gamma = \sum_{i=1}^m \gamma_i \sin \omega_i t$

Linearity condition
 $\gamma_1 + \gamma_2 + \gamma_3 + \dots + \gamma_{10} < LVE \text{ limit}$



Next Class

More on fractals – what is it?

Real data – gelation and gel point

Physical vs chemical gels

Thixotropy